

Vishal Kumar Rai

Ph.D. Candidate, Economics — Structural Macroeconomics & Life-Cycle Modeling
Indian Statistical Institute – Delhi

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Summary

Ph.D. candidate in Economics at the Indian Statistical Institute, Delhi, specializing in structural macroeconomics and life-cycle modeling. Research focuses on retirement, health, and labor supply under uncertainty, with applications to environmental, epidemiological, and occupational risks. Uses dynamic programming, continuous-time optimization, and microdata calibration for structural policy analysis.

Research Papers

- ***Climate, Pollution and Endogenous Retirement***: Dynamic life-cycle model linking pollution, health investment, and retirement; calibrated to microdata for policy analysis.
- ***Endogenous Retirement under a Pandemic***: Structural model of mortality risk and health expenditure with retirement choice under uncertainty.
- ***Occupational Variation in Work Strain***: Life-cycle model with occupational heterogeneity in labor disutility and retirement timing.

Computational Tools

SwitchingControl.jl — Registered Julia package for endogenous regime-switching ODE systems using event-driven solvers, designed for macroeconomic and optimal control applications.

Research Interests

Macroeconomic Policy • Public Finance • Social Protection • Climate and Health Economics
• Computational Methods • Life-Cycle Modeling

Education

Ph.D. in Economics, Indian Statistical Institute – Delhi *2019 – Present (Expected 2026)*
Thesis: Essays on Life-Cycle Decisions under Environmental and Epidemiological Risks
M.A. in Economics, Delhi School of Economics *2016 – 2019*
B.A. (Hons.) Business Economics, University of Delhi *2012 – 2015*

Skills

Programming: Julia, Python **Methods:** Dynamic Programming, Structural Modeling
Data: MEPS, CPS, Life Tables **Tools:** L^AT_EX, Git

Experience & Teaching

Teaching Assistant, ISI Delhi (Prof. Chetan Ghate)
Macroeconomics I, Global Macroeconomics (MSQE and Ph.D. coursework)

- Conducted tutorial sessions and problem-solving discussions
- Assisted with grading, evaluation, and course coordination

Awards

Junior Research Fellowship (JRF), ISI (2019) Top 25 Rank – DSE Entrance (2016)
Academic Excellence – CVS 3rd Rank (2014) CBSE Merit Certificate – 10 CGPA (2010)